The SPY Breakeven Volatility Surface

Executive Summary | Block-Bootstrap Delta-Hedge Simulation vs. Market Implied Vols

What We Built

We estimate SPY's **breakeven volatility** -- the hedging volatility that makes a delta-hedged option position's P&L exactly zero over a realized price path -- using a block bootstrap of 10 years of real SPY history. We then compare it against SPY's live market-implied volatility surface to quantify the **variance risk premium (VRP)**: how much more the market charges for options than history says was actually needed.

Methodology

1. Delta-hedge one option along one simulated path; root-find the vol σ that makes total hedge P&L exactly zero.
2. Block-bootstrap 1,000+ paths (21-trading-day blocks drawn from 10 years of real history) → a distribution of breakeven vols per option; we report the mean.
3. Repeat for 80% put / 90% put / 100% call / 110% call, all against the same path ensemble → a breakeven volatility **smile**.
4. Expand to 8 moneyness levels (80%-115%) × 7 expiries (1M-3Y) = 56 cells → a full breakeven volatility **surface**.
5. Pull SPY's live option chain, matched onto the same grid, and compute VRP = Implied Vol – Breakeven Vol.

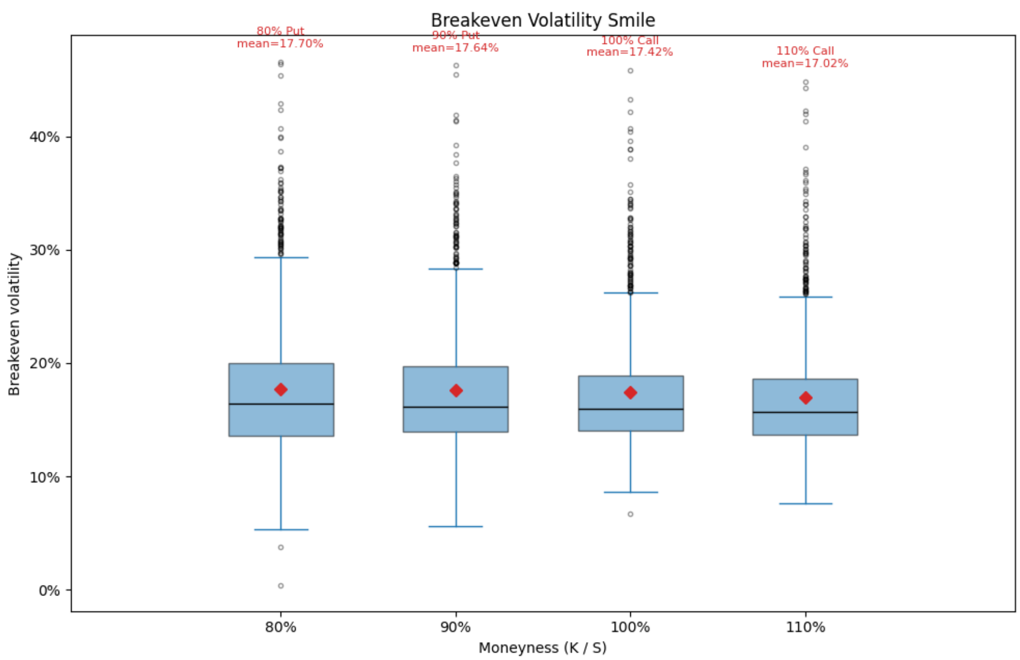
Breakeven Condition

Total $P\&L(\sigma) = \Sigma_t \left[C(S_{t+1}) - C(S_t) - \Delta_t \cdot (S_{t+1} - S_t) \right] \to$ solve for σ_{BE} such that Total $P\&L(\sigma_{BE}) = 0$

Key Result: The Breakeven Volatility Smile (1-Year)

Moneyness	80% Put	90% Put	100% Call	110% Call
Breakeven Vol	17.70%	17.64%	17.42%	17.02%

Breakeven vol declines monotonically moving from the 80% put to the 110% call — the **leverage effect**: equity selloffs historically coincide with higher realized volatility, so a path that drifts toward the 80% strike is more likely to have passed through a higher-vol regime than one drifting toward the 110% strike.



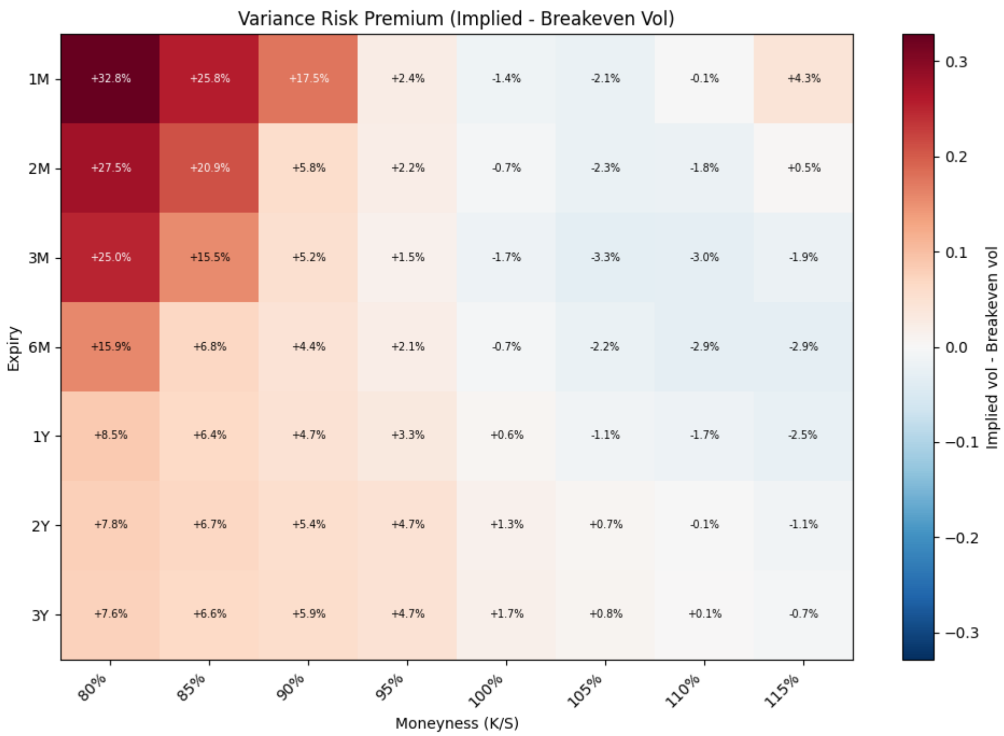
The Full Surface

The same procedure run across all 56 (expiry, moneyness) cells produces a full breakeven vol surface. **Caveat:** at very short-dated, deep out-of-the-money combinations (e.g. 1M at 80% moneyness), the option is worth effectively zero at any realistic volatility, so no meaningful breakeven vol exists there. Our pipeline detects this (P&L ≈ 0 across the entire search range) and reports it as missing rather than fabricating a number from numerical noise.

Comparison to the Market: The Variance Risk Premium

VRP = $\sigma_{IV} - \sigma_{BE}$ (vol-point terms), or $\sigma_{IV}^2 - \sigma_{BE}^2$ (textbook variance terms). Positive VRP means the market charges more for an option than historical hedging outcomes say was needed — compensation for tail/crash risk that a 10-year realized-vol sample doesn't fully capture.

Finding: VRP is largest for short-dated, deep out-of-the-money puts (+32.8 vol points at 1-month / 80% moneyness) and decays toward roughly zero further out in time and moneyness. Three drivers: (1) crash/tail-risk premium market makers charge beyond what history justifies, (2) persistent institutional demand for portfolio insurance (structural put buying), and (3) a supply-demand imbalance — few natural sellers of crash insurance, so those who do demand compensation for the risk they absorb.



Bottom Line

SPY options price in a substantial, measurable variance risk premium concentrated in short-dated downside protection — consistent with the market charging more for crash insurance than a decade of realized history says was strictly necessary.

Limitations: *breakeven-vol distributions are right-skewed (mean > median) and we report the mean; the block bootstrap only resamples the observed ~10-year history and cannot produce a crash unlike anything already in the sample; market matching snaps to the nearest listed strike/expiry, adding a small amount of basis noise.*